RAJENDRAN K

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Professional Summary

Software Engineer with hands-on experience building and optimizing cloud-native distributed systems using Java, Spring Boot, AWS, and Kubernetes. Experienced in developing secure backend services, optimizing application performance, and deploying scalable applications on AWS. Inventor of a published traffic management patent and author of an AI-focused Springer publication. Passionate about building reliable software while leveraging AI-assisted development tools such as Codex and Claude to enhance engineering productivity, targeting SDE roles requiring system-level thinking and engineering ownership.

Technical Skills

Languages	Java, Go, Python, SQL, Bash
Backend & Distributed Systems	Spring Boot, RESTful APIs, RabbitMQ, NATS, Hibernate/JPA, HikariCP
Cloud & Infrastructure	AWS (EKS, IAM, KMS, EventBridge, EC2), Docker, Kubernetes, Karpenter, Streamlit
Databases & Storage	MySQL, Oracle, MariaDB, MongoDB, Redis, Dragonfly, Guava, RocksDB
Observability & Reliability	Prometheus, Grafana, cAdvisor, Structured Logging, Hey (Load Testing)
Development Tools	Git, Linux CLI, Postman, DBBeaver, IntelliJ IDEA, VS Code, Jupyter Notebook

Work Experience

Software Engineering Intern (SDE) — **Oct 2025 – Jun 2026**
Tanla Platforms Pvt. Ltd. — *Hyderabad, India*

- Implemented a two-tier caching architecture **(L1: Google Guava in-memory cache; L2: DragonflyDB)** reducing Guava memory overhead by ∼**40%** and lowering overall component memory footprint.
- Engineered a centralized **Key Management System** using AWS KMS with automated key rotation and secure DEK handling across communication platforms, reducing key management effort by ∼**60%** while improving secure data retrieval by ∼**40%**.
- Enchanced the **per-sender ThreadPoolExecutors** in the Dispatcher Microservice for concurrent message processing across multiple Sender IDs, achieving **242 avg TPS**(**21% above** the production target) with **zero message loss** on graceful restarts across 50K load tests.
- Validated and deployed **AWS EKS autoscaling** using Karpenter through 10+ production scenarios involving Spot interruption handling and node provisioning strategies, enabling infrastructure cost savings of ∼**65%**.
- Performed large-scale storage benchmarking on **2 million+ records** using RocksDB, evaluating WAL configurations, read/write throughput, and latency, preventing a high-risk production migration through performance-driven analysis.
- Built a **Go HTTP connection pool POC** integrated with cAdvisor, Prometheus, Grafana for observability across 10+ configurations, demonstrated superior memory, CPU, and throughput performance over JDK 8 and JDK 21, supporting Go adoption for high-traffic microservices.

Projects

Cloud-Based Malware Detection System | **Link** — *Jan 2025 – May 2025*

- Developed a malware detection application using **Scikit-learn & TensorFlow**, achieving **98% accuracy** in classifying malicious and legitimate files.
- Deployed the application on **Streamlit Cloud** with automated CI/CD workflows, enabling real-time predictive file scanning; integrated **AES & DES encryption** for data security and audit controls.

Round-Key Adaptive Cipher (RKAC) Algorithm — *Jul 2024 – Nov 2024*

- Designed a novel symmetric encryption algorithm with **adaptive key expansion, dynamic S-Box**, and unique round transformations, enhancing resistance against cryptanalysis by ∼**40%** compared to baseline AES.

IoT-Based Smart Collar for Real-Time Tracking & Accident Prevention — *Jul 2024 – Nov 2024*

- Created an **ESP8266 + GPS + LoRa** smart collar with geofencing, reducing cattle-road accident risk by ∼**35%** in trials; devised a Flutter dashboard with instant alerts, cutting owner response time by ∼**40%**.
- Integrated **Firebase & ThingSpeak** cloud storage for secure monitoring, increasing data accessibility by ∼30%.

Secure QR-Based Identity Management System — *Jul 2022 – Nov 2022*

- Developed a Java-based identity management system with secure QR authentication, improving ID generation efficiency by ∼**30%**. Implemented **AES-256 encryption**, role-based authentication, password-protected PDFs, and SQL injection prevention to strengthen application security and data protection.
- Formulated backend modules using **Java Swing, JDBC, & MariaDB**, integrating secure database operations and input validation by ∼**20%**.

Education

Vellore Institute of Technology — **Jun 2021 – May 2026**
Int M.Tech Software Engineering | CGPA: 9.12/10.00 — *Vellore, India*